

HVIDE SANDE, Denmark

WMO: 060580

Lat: 56.007N

Lon: 8.141E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.0	-4.9	-9.4	1.7	-6.1	-7.3	2.0	-3.8	17.3	3.4	15.5	2.8	6.1	60	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	24.6	19.6	22.7	18.7	21.1	17.8	20.3	23.5	19.2	22.1	18.2	20.6	4.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.2	17.9	12.9	20.8	17.1	12.2	19.9	58.1	23.7	54.5	22.3	51.4	20.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.4	12.1	-8.0	27.5	4.2	2.2	-11.0	29.1	-13.4	30.5	-15.8	31.7	-18.8	33.4		
(5)			-8.5	21.7	4.0	1.4	-11.3	22.7	-13.7	23.5	-15.9	24.3	-18.8	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	2.2	1.9	3.7	7.6	11.4	14.2	17.1	17.5	14.7	10.9	6.7	3.9
	DBStd	6.22	3.57	3.26	2.66	2.83	3.04	2.58	2.76	2.14	2.00	2.45	3.00	3.78
	HDD10.0	1074	242	227	194	83	16	1	0	0	0	20	102	189
	HDD18.3	3313	501	461	452	321	215	127	58	42	109	232	348	448
	CDD10.0	847	0	0	0	12	61	127	221	232	142	47	4	0
	CDD18.3	44	0	0	0	0	1	4	21	16	2	0	0	0
	CDH23.3	116	0	0	0	0	3	9	68	35	1	0	0	0
	CDH26.7	11	0	0	0	0	0	1	7	3	0	0	0	0
(14) Wind	WSAvg	6.5	7.4	6.9	6.6	6.1	6.1	6.2	5.7	6.0	6.5	7.0	7.1	6.9
(15) Precipitation	PrecAvg	815	69	50	47	37	43	61	60	84	88	99	87	80
	PrecMax	1018	114	95	101	68	77	143	100	147	170	208	176	176
	PrecMin	572	11	11	12	6	3	13	13	22	34	27	26	22
	PrecStd	123	27	24	26	16	20	36	24	32	38	47	34	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	7.9	10.8	17.9	23.0	24.6	27.8	26.7	22.3	16.7	13.6	10.5
		MCWB	7.9	6.6	8.1	13.0	17.4	18.3	21.0	20.7	18.6	15.2	12.4	9.3
	2%	DB	8.1	7.1	8.6	15.6	20.3	21.9	25.3	24.0	19.8	15.6	12.4	9.5
		MCWB	6.8	6.1	6.8	11.8	15.6	17.2	20.2	19.6	16.9	14.0	11.1	8.5
	5%	DB	7.3	6.4	7.5	13.7	18.2	19.8	23.2	22.0	18.5	14.7	11.5	8.9
		MCWB	6.2	5.5	6.2	10.7	14.2	16.4	19.2	18.6	16.0	13.3	10.2	7.9
	10%	DB	6.6	5.8	6.7	11.8	16.1	18.0	21.2	20.6	17.5	14.1	10.6	8.3
		MCWB	5.6	5.0	5.7	9.5	13.1	15.4	18.1	17.8	15.3	12.7	9.2	7.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	6.9	8.2	13.7	18.2	19.5	22.4	21.7	19.0	15.5	12.7	9.7
		MCDB	8.9	7.5	9.9	16.7	21.6	23.3	26.0	24.5	21.7	16.5	13.3	10.2
	2%	WB	7.1	6.2	7.2	12.3	16.1	17.9	20.7	20.1	17.5	14.5	11.5	8.8
		MCDB	7.7	6.7	8.1	15.0	19.6	20.6	24.1	23.2	19.2	15.2	12.2	9.3
	5%	WB	6.5	5.7	6.5	11.0	14.7	16.8	19.5	19.0	16.5	13.7	10.4	8.2
		MCDB	7.1	6.2	7.2	13.3	17.7	19.4	22.6	21.7	18.0	14.4	11.2	8.8
	10%	WB	5.7	5.2	5.9	9.7	13.2	15.6	18.4	18.1	15.7	13.0	9.5	7.5
		MCDB	6.4	5.7	6.6	11.4	15.7	17.8	21.1	20.2	17.2	13.9	10.3	8.1
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	2.8	3.3	4.3	4.3	3.8	4.1	3.9	3.6	3.1	2.9	3.0
		MCDBR	2.6	2.5	4.0	7.3	8.0	7.4	7.8	6.4	5.1	3.5	2.7	2.4
		MCWBR	2.9	2.3	2.9	4.5	4.8	4.6	4.0	3.5	3.3	3.0	3.0	2.7
	5% WB	MCDBR	2.5	2.4	3.6	7.0	7.4	6.6	7.1	5.7	4.6	3.3	2.6	2.4
		MCWBR	2.9	2.3	2.9	4.5	4.9	4.5	4.1	3.6	3.5	3.1	3.0	2.7
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.362	0.395	0.385	0.382	0.394	0.397	0.377	0.356	0.333	0.309
	taud		2.229	2.291	2.313	2.269	2.367	2.398	2.379	2.375	2.412	2.384	2.289	2.228
	Ebn at Noon		559	712	797	821	856	862	842	812	776	690	548	481
	Edh at Noon		57	78	96	116	112	110	110	104	88	71	54	45
(44) All-Sky Solar Radiation	RadAvg		0.46	1.06	2.42	4.06	5.36	5.70	5.31	4.24	2.86	1.45	0.57	0.33
	RadStd		0.08	0.15	0.23	0.45	0.51	0.46	0.54	0.32	0.28	0.17	0.09	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	-130	-182	N/A

Nomenclature: See separate page